

Project Background

- The client, for whom this device is designed, has multiple sclerosis (MS), which is currently in remission. MS is a disease in which the body attacks the nerves. Due to this, he has significant difficulty lifting his legs more than an inch or two from the ground. This makes getting in and out of vehicles, into bed, and putting his feet on a stool very difficult.
- The client currently uses a cane to ambulate, and wanted to continue to use a cane without needing a second device to move and lift his legs.
- The project is about creating a device that can help the client lift his legs into and out of vehicles, onto stools, on a stool ...etc. without compromising his stability and is operable by a single person.

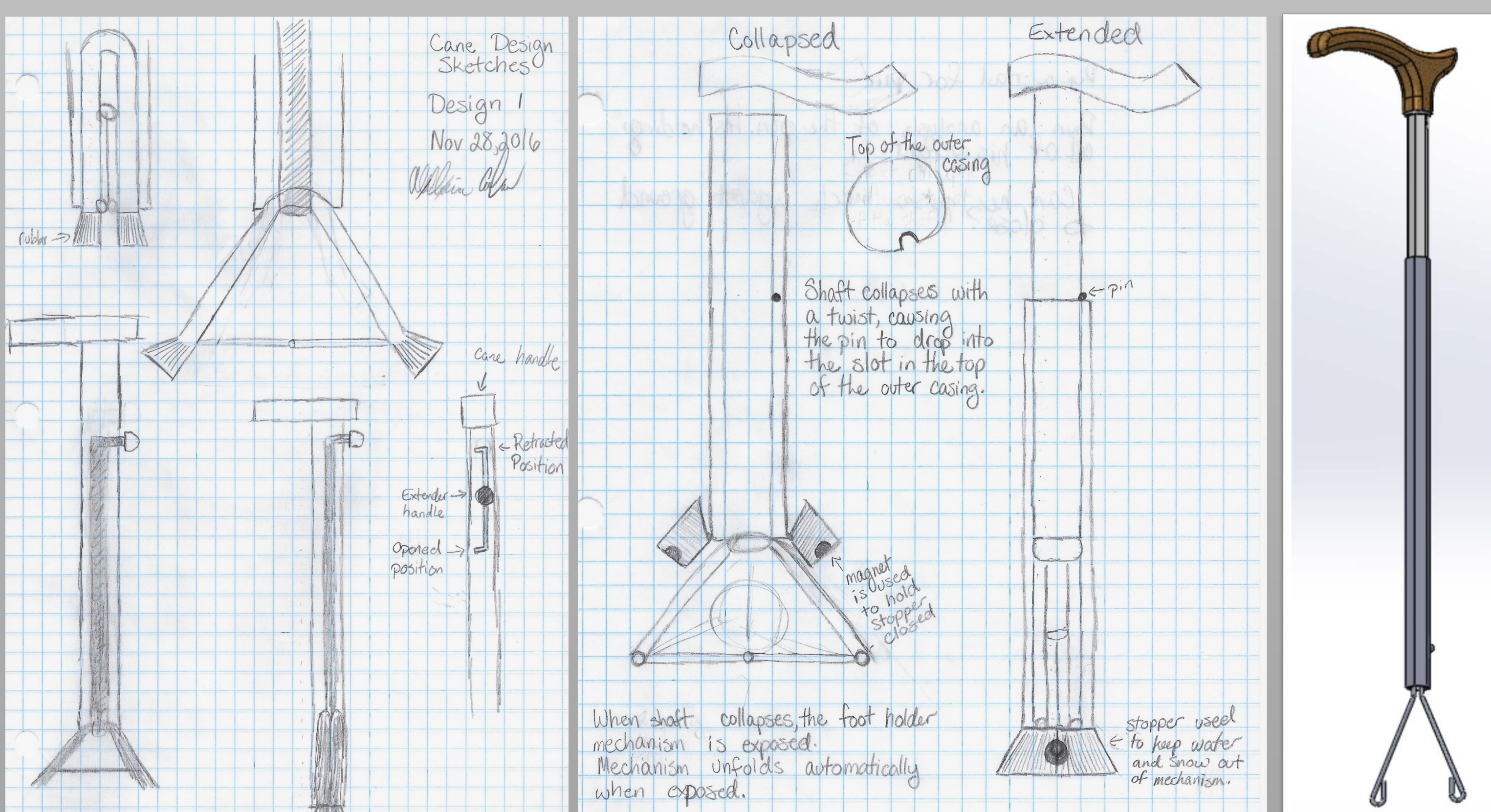
Current Solutions



Leg lifters are relatively cheap and common items which are marketed for people who have difficulty raising their legs or who have a cast. They are made of two nylon loops and sometimes have a stiff portion in the middle of the device. They are often too long and can be difficult to get around the foot.

Project Design Parameters

- Must be one device which incorporates a cane
- Must be durable, the device is intended for day to day use.
- Must be 35 inches tall.
- Must be no longer than 24 inches when being used as a leg lifter so it is able to be used in a vehicle setting
- Must be light weight, < 1 lb



Early sketches of the design



CAD drawing of the design

Prototypes

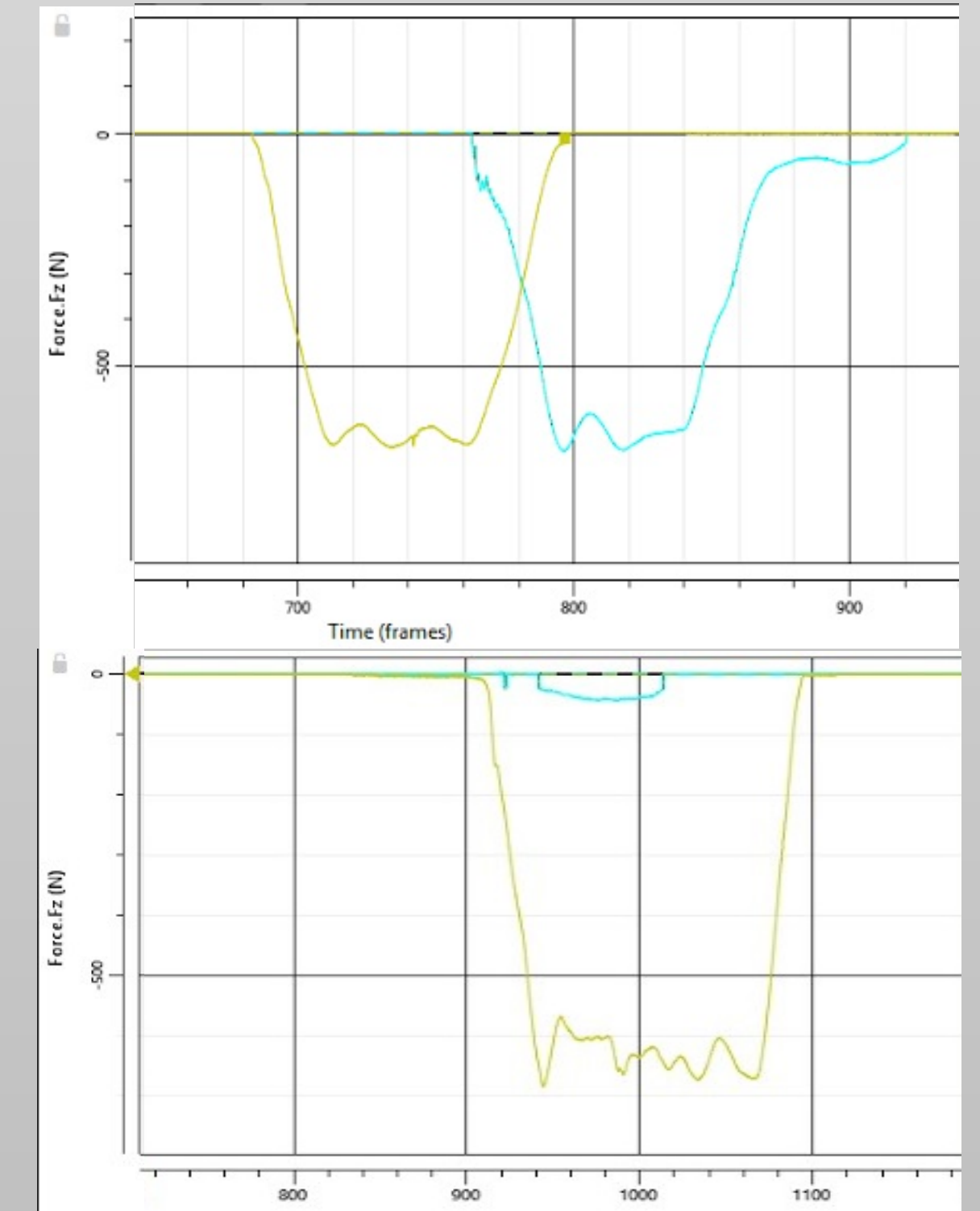
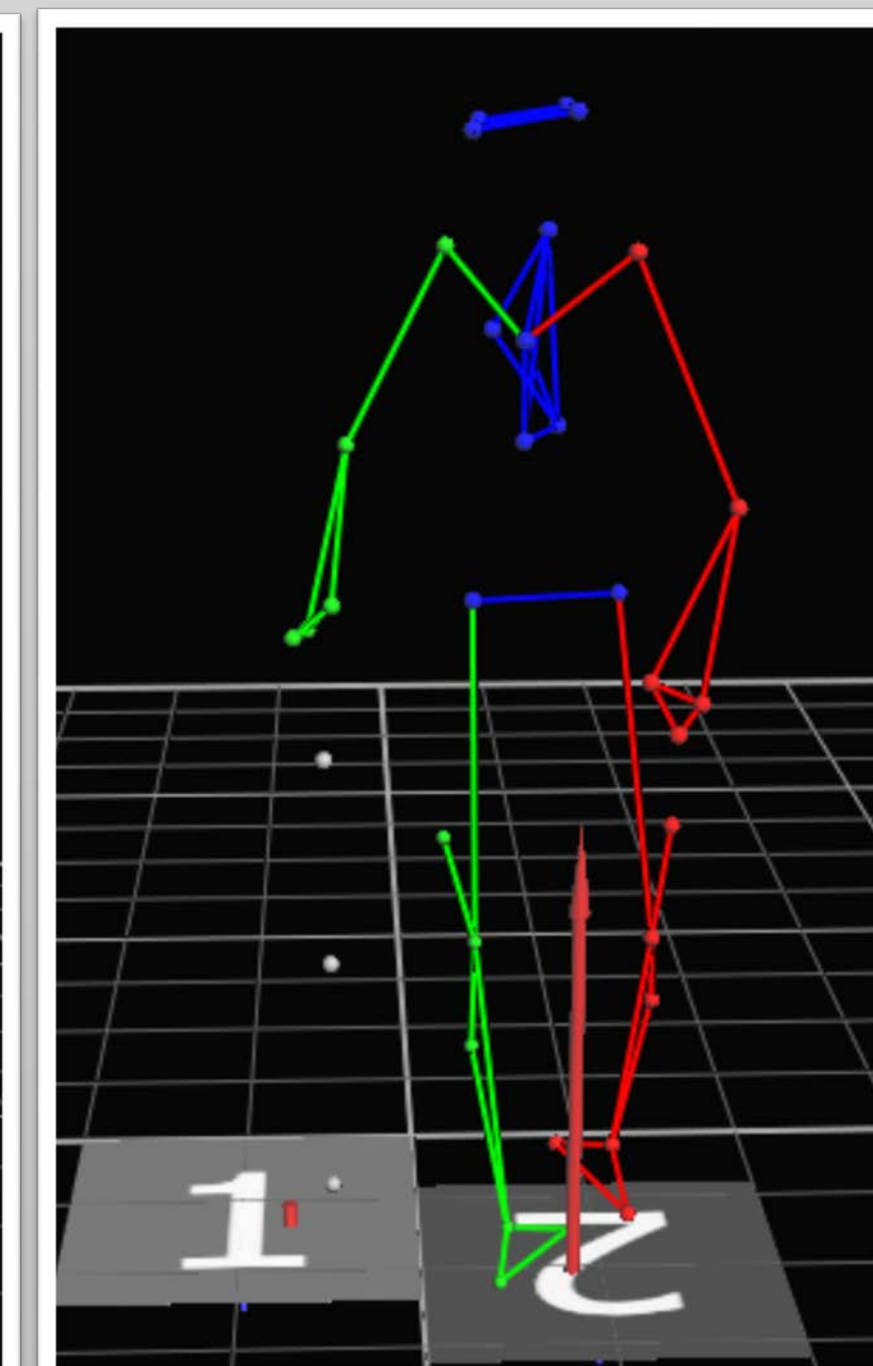
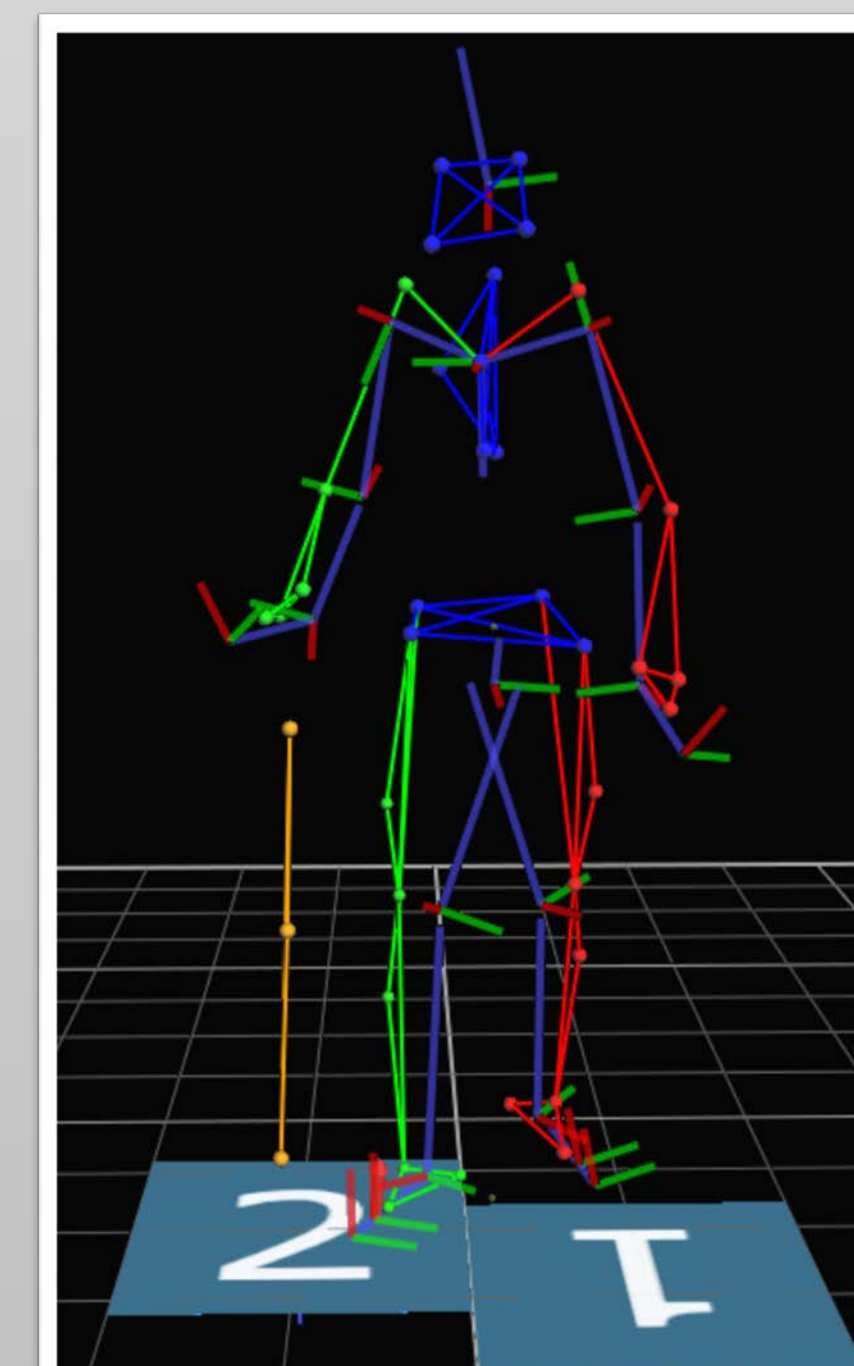


Above: The first mock up of the cane of PVC, coat hanger, wire, and scrap wood.

Right: The second prototype of carbon fiber, 3D printed handle, wire, nylon, rubber tip

Validation and Testing

- To ensure the cane will not change the client's gait
 - Vicon Motion Capture system and Kistler Force Plates
 - Control data with a regular cane vs. Walk&Lift Product testing with the client
 - The client was asked to walk normally with the cane, and was asked to stand from a seated position with the help of the cane. Three trials were recorded for each test.



Testing setup and results

- To ensure the cane will support the client's weight
 - Instron compression testing
 - Failure at 4700 lb.
- To ensure safety
 - Failure Modes and Effects Analysis (FMEA)
 - Hazards Analysis (HA)

Future Considerations

- Using a mold for the cane handle instead of fabricating each one individually
- Fabricating a cane with aluminum and comparing it to carbon fiber

Acknowledgements

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